Q. 1 (Tiger-ants)**.** Two ants exit their eating club onto Prospect Ave. after a long evening of partying. They are a bit disoriented, and each ant starts walking according to an independent standard Brownian motion. (The eating club is at position 0.)

a. Given that the first ant is at location x at time s, what is her (conditional) mean square distance from the eating club at time $t > s$?

b. What is the probability that at time t, the distance between the two ants is at most r? (You may express your answer as an integral.)

Q. 2 (Fruity yogurt in the COVID era)**.** In order to help society in its time of need, the Fruity Yogurt store has decided to reprogram its bubble tea machine to produce hand sanitizer instead. Due to an unfortunate labeling mishap, however, not all customers are aware of this change. Customers arrive at the store according to a Poisson process with an average rate of 10 customers per hour. Each customer helps him/herself to the bubble tea machine with probability 0.4; among those customers who use the machine, each has a probability 0.25 that they fail to read the posted notice on the machine and try to drink the hand sanitizer. All customers are independent of each other.

a. What is the expected amount of time it takes until the first time a customer drinks the hand sanitizer? What is the variance of this time?

b. Given that the second time a customer drinks the hand sanitizer happens exactly at time $t = 1$ hour, what is the (conditional) expectation of the first time that a customer drinks the hand sanitizer?

Q. 3 (Rocky Horror Picture Show)**.** In this cult classic, there is a song that goes: "It's just a jump to the left. . . And then a step to the right. . . Let's do the time warp again!" The writers of the song are said to have been deeply inspired by a particular kind of random walk. In each step, the dancer makes a jump to the left, a jump to the right, or a time warp with equal probability. A time warp is two jumps to the right. A jump to the left decreases your position by 1, and to the right increases your position by 1.

The criminologist who demonstrates the dance in the movie dances on a table. The table is only 3 jumps wide: it consists of positions $0, 1, 2$. If you move beyond that you fall off the table. The criminologist starts his dance at position 1.

a. What is the probability that the criminologist will fall off the table in at most two dance steps?

b. What is the expected number of steps until the criminologist falls off the table?